9-10-25
Exp. 8

Long-Short Term Memory (LSTM)

Aim:

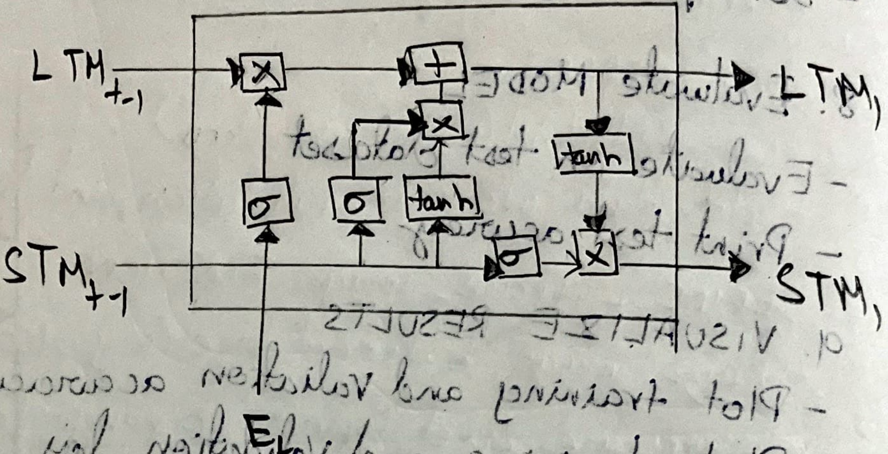
To develop a long-short term memory based Recurrent Neural Network to classify titanic passengers based on survival probability and assess its accuracy

Objective:-

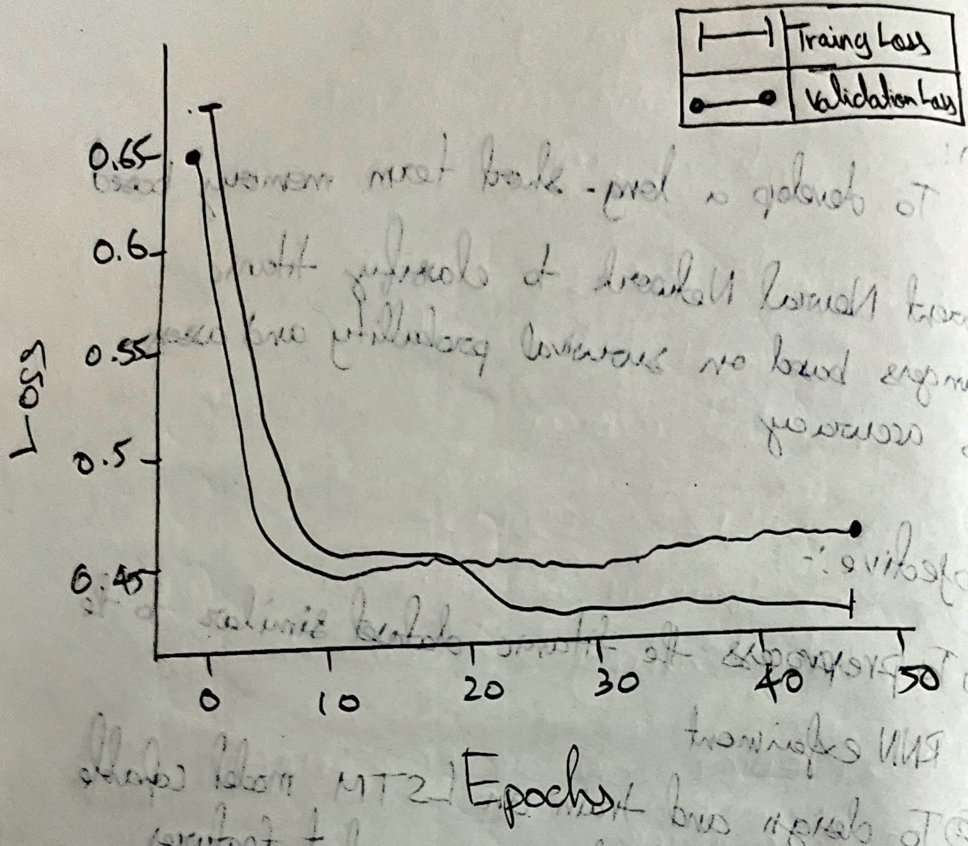
1. To preprocess the titanic dataset similar to the RNN experiment
2. To design and train an LSTM model capable of capturing dependencies in input features
3. To compare training and validation performance visually using loss and accuracy plots
4. To calculate and print final validation accuracy for model evaluation

PseudoCode:-

1. Load the titanic dataset
2. Select features: Pclass, Sex, Age, Fare and the target variable survived
3. Fill missing Age values and encode the sex feature numerically
4. Standardize features and reshape them into 3D format for LSTM input.



LSTM Architecture



Observation :-

Epoch 10, Loss : 0.4053

Epoch 20, Loss : 0.4039

Epoch 30, Loss : 0.4023

Epoch 40, Loss : 0.4015

Epoch 50, Loss : 0.3997

Accuracy : 80.45%

⑤ Split data into training and validation sets

⑥ Define a Sequencial model

a. Add an LSTM layer with 32 units and 'tanh' activation

b. Add a Dense output layer with 1 neuron and 'sigmoid' activation

⑦ Compile the model with 'adam' optimizer and 'binary_crossentropy' loss

⑧ Train the model for 50 epochs with batch size 16 and validate on valiation data

⑨ Plot training vs validation loss and accuracy curves

⑩ Evaluate the model and print final validation accuracy.

Result :-

The code executed succesfully with an accuracy of 80.45%. and the Loss vs Epoch curve was obtained.